

Practice Quiz: Action

Part A: Multiple Choice

1. In Active Inference, action arises from: A) A central executive that issues motor commands B) Prediction error minimization — the body moves to fulfill proprioceptive predictions C) Random motor output filtered by natural selection D) Conscious deliberation followed by voluntary effort
2. Gibson's concept of affordances refers to: A) The financial cost of actions B) Action possibilities that the environment offers to a particular organism C) The neural pathways dedicated to motor control D) Abstract representations of possible actions in working memory
3. Expected Free Energy (EFE) decomposes into: A) Accuracy and complexity B) Pragmatic value (risk) and epistemic value (ambiguity resolution) C) Motor planning and sensory prediction D) Conscious and unconscious motivation
4. An infant shaking a rattle to learn about its properties is best described as: A) Random motor babbling with no cognitive function B) Epistemic action — acting to reduce expected ambiguity C) Pragmatic action — seeking the reward of the sound D) A reflex with no inferential component
5. Active Inference's account of free will is best classified as: A) Hard determinism — actions are fully determined and freedom is an illusion B) Libertarianism — the agent has irreducible causal power beyond physical processes C) Compatibilism — freedom is consistent with deterministic inference over a flexible generative model D) Mysterianism — free will is beyond human comprehension
6. Dreyfus's phenomenology of expertise describes experts as: A) Highly deliberate, step-by-step thinkers B) Absorbed copers whose skilled actions flow without explicit deliberation C) Unable to explain their performance because expertise is non-cognitive D) Reverting to rule-following under pressure

7. The difference between exploratory and exploitative behavior in Active Inference corresponds to: A) The difference between perception and action B) The balance between epistemic value (information gain) and pragmatic value (preference satisfaction) C) The distinction between conscious and unconscious processes D) The difference between biological and artificial agents

Part B: Short Answer

1. Explain Friston's claim that "I don't move my arm because I want to; I move my arm because I predict I will." What is the philosophical significance of this inversion?
2. A robot that minimizes prediction error in a warehouse performs the same functional operations as an Active Inference agent. Does it "act" in a philosophically meaningful sense? Defend your answer.
3. Construct a brief argument either supporting or challenging the claim: "Moral responsibility requires that the agent could have done otherwise, but Active Inference suggests that the agent always does what its generative model dictates."